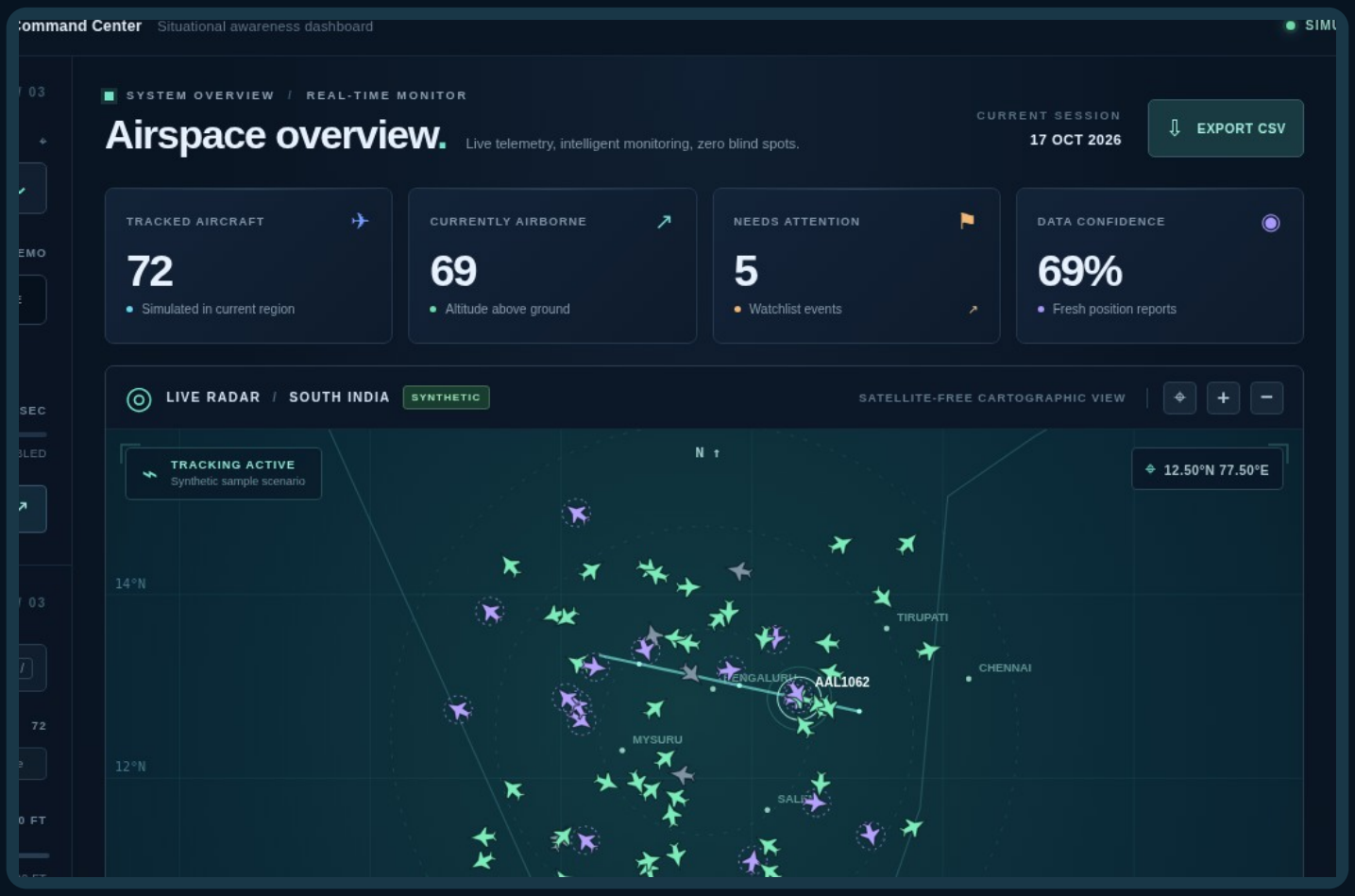
AIRSPACE INTELLIGENCE

Track the sky.

Know what you can trust.

Not just aircraft dots: a resilience-first telemetry observability cockpit.



01 MONITOR

OpenSky state vectors and six regions

02 VERIFY

Fresh / delayed / stale confidence

03 EXPLORE

Local replay and 60s estimates

04 RESPOND

Custom geofence + proximity watch

TECHNICAL SHOWCASE

Resilient by design.

Live data is useful only when its freshness and limitations are visible.

LIVE PIPELINE

- > Browser UI -> FastAPI proxy -> OAuth2 / OpenSky
- > Server-side regional cache, request coalescing, daily quotas
- > Graceful 429 handling, backoff, browser snapshot fallback

DISTINCTIVE FEATURES

- > Confidence-scored reported positions vs. purple estimates
- > Replay of locally collected observations (not official history)
- > Experimental proximity watch and circular geofence rules
- > Offline, clearly labeled synthetic flight scenario

ENGINEERING QUALITY

- > Interactive Canvas map with coastline geometry and pan/zoom
- > Browser-native JS modules + Python FastAPI (no npm build)
- > 14 passing automated tests, Docker, GitHub Actions CI

60-SECOND INTERVIEW PITCH

SkySentinel turns imperfect OpenSky aircraft data into a transparent dashboard.
It distinguishes live reports, cached states, time-based estimates and synthetic demo data, while remaining interactive if the upstream API fails.
The extra layer is a confidence-aware radar with replay and configurable watches.

STATUS: Source code and tests are ready; live API must be validated on your network.
Public deployment requires your own hosting account; Docker/Render config included.